

Device Certificate Creation Steps(CVIP Preprod)

1. Create Folder & Copy CA Files

```
$ mkdir device_cert && cd device_cert
```

2. Generate Private Key

```
$ openssl genpkey -algorithm RSA -out device_private.key -pkeyopt rsa_keygen_bits:2048
```

3. Create CSR

```
$ openssl req -new -key device_private.key -out device.csr \
-subj
"/C=IN/ST=State/L=City/O=Traxo/OU=Software/CN=355013421200005-B234ZTZT03097"
```

4. Create Payload

```
$ TS=$(date +%s)
$ csr_data=$(awk 'BEGIN{ORS="\n"} {print}' device.csr)

cat > payload.json <<EOF
{
  "commonName": "355013421200005-B234ZTZT03097",
  "timeStamp": $TS,
  "csr": "${csr_data}"
}
EOF
```

5. Call Certificate API

```
$ TOKEN="d038a290-8fc0-30bf-9390-6d9c02317f6f"

$ curl -v --tlsv1.2 \
-H "Authorization: Bearer ${TOKEN}" \
-H "Content-Type: application/json" \
-d @payload.json \
"https://lb1.cvip-preprod.citroen.in:40543/csr/createCertificate" -o
device_cert_response.json
```

6. Extract Certificate

```
$ jq -r '.message' device_cert_response.json > device_cert.b64
$ base64 -d device_cert.b64 > device_cert.der
```

```
$ openssl x509 -inform der -in device_cert.der -out device_cert.pem
```

7. Verify Certificate

```
$ openssl x509 -in device_cert.pem -noout -subject -issuer  
$ openssl verify -CAfile cvip_ca_bundle.pem device_cert.pem
```

8. Build Device Joined Payload

```
$ VIN="B234ZTZT000003097"  
$ IMEI="355013421200005"  
$ TBOX_SN="B234ZTZT03097"  
$ TS=$(date +%s%3N)  
  
$ CERT_B64=$(openssl x509 -in device_cert.pem -outform der | base64 -w0)  
  
$ cat > device_join_payload.json <<EOF  
{  
  "vehicleId": "$VIN",  
  "TboxSerialNum": "$TBOX_SN",  
  "imeiNo": "$IMEI",  
  "protocolVersion": "2.0.0",  
  "TboxOperatingState": "NORMAL",  
  "TboxApplicationState": "FACTORY",  
  "TboxeSimState": "NORMAL_SIM",  
  "CCPUVersion": "MD0.00.01",  
  "VMCUVersion": "MD0.00.01",  
  "timestamp": $TS,  
  "certificate": "$CERT_B64"  
}  
EOF
```